

Sardar Vallabhbhai National Institute of Technology, Surat
Computer Science and Engineering Department
M.Tech(CSE/DS)-I - Sem-I - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Exploring CVIP Tools

Problem Statement:

Each student is assigned two freely available tools related to Computer Vision, Image Processing, 3D Vision, or Medical Image Analysis. Explore the assigned tools for one week and perform a small, meaningful task using each tool.

Student	Tool 1	Tool 2
P26CS014	Ultralytics YOLO	Fiji
P26CS018	Open3D	CloudCompare
P26CS026	napari	3D Slicer
P26DS001	MeshLab	Meshmixer
P26DS015	nerfstudio	Blender
P26DS016	Tinkercad	3DSlash
P26DS017	Spline	Womp
P26DS022	Vectary	SketchUp Free
P26DS026	Figuro	Clara.io

Task:

For each assigned tool:

- Understand its purpose and identify one CV/IP-related application.
- Perform one small practical experiment.
- Show the input and output of your experiment.
- Identify the CV/IP concept involved and briefly explain it.
- Note one observation or limitation.

One-Week Presentation:

After one week, each student will demonstrate both tools and briefly explain:

- What the tool does.
- What experiment was performed.
- What CV/IP concept is involved.
- Input, output, and key observation.

Duration: 5–7 minutes per student.

Important: The objective is not to learn all features of the tools. The focus is on understanding a CV/IP concept through practical exploration.